

Tutorial 6 — Variance of the OLS Estimator and Hypothesis Testing

Questions

Introductory Econometrics — TA Session

Duration: 50 minutes

Based on Wooldridge, Theorem 2.2, Sections 2.5 and 4.1–4.2

Block 1: Motivation — Why Isn't Unbiasedness Enough? (3 min)

The Simple Linear Regression Assumptions

Label	Name	Statement
SLR.1	Linear in Parameters	$y = \beta_0 + \beta_1 x + u$
SLR.2	Random Sampling	$\{(x_i, y_i)\}_{i=1}^n$ are i.i.d. draws
SLR.3	Sample Variation in x	$SST_x = \sum (x_i - \bar{x})^2 > 0$
SLR.4	Zero Conditional Mean	$E[u \mid x] = 0$
SLR.5	Homoskedasticity	$\text{Var}(u \mid x) = \sigma^2$ (constant)
SLR.6	Normality	$u \mid x \sim N(0, \sigma^2)$

Under **SLR.1–SLR.4**: $E[\hat{\beta}_1] = \beta_1$ (unbiasedness). Adding **SLR.5**: we get $\text{Var}(\hat{\beta}_1)$. Adding **SLR.6**: exact t -distribution for hypothesis tests.

Theorem 2.2 (Wooldridge) — Under SLR.1–SLR.5:

$$\boxed{\text{Var}(\hat{\beta}_1 \mid \mathbf{x}) = \frac{\sigma^2}{SST_x}} \quad \text{where} \quad SST_x = \sum_{i=1}^n (x_i - \bar{x})^2$$

The proof relies on a key representation of $\hat{\beta}_1$. Starting from the OLS formula and substituting $y_i = \beta_0 + \beta_1 x_i + u_i$ (**SLR.1**):

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{SST_x} = \frac{\beta_1 SST_x + \sum (x_i - \bar{x}) u_i}{SST_x}$$

where we used $y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (u_i - \bar{u})$ and the fact that $\sum (x_i - \bar{x}) = 0$ eliminates

$\bar{u}$. This gives:

$$\hat{\beta}_1 = \beta_1 + \sum_{i=1}^n w_i u_i, \quad w_i = \frac{x_i - \bar{x}}{\text{SST}_x}$$

$\hat{\beta}_1$ equals the true parameter plus a weighted sum of the errors. The weights w_i depend only on the x -values and are constants conditional on $\mathbf{x}$.

Block 2: Deriving and Computing $\text{Var}(\hat{\beta}_1)$ (12 min)

Question 1 (Derivation and numerical computation)

- (a) Starting from the representation $\hat{\beta}_1 = \beta_1 + \sum_{i=1}^n w_i u_i$, derive $\text{Var}(\hat{\beta}_1 \mid \mathbf{x}) = \sigma^2 / \text{SST}_x$.

Hint: Follow these steps — (i) Why can you drop β_1 from the variance? (ii) Expand $\text{Var}(\sum w_i u_i \mid \mathbf{x})$. Which assumption eliminates the covariance terms? (iii) Which assumption makes $\text{Var}(u_i \mid \mathbf{x})$ the same for all i ? (iv) Simplify $\sum w_i^2$.

- (b) Suppose $n = 5$ observations with x -values $\{1, 3, 5, 7, 9\}$ and $\sigma^2 = 10$. Compute $\bar{x}$, SST_x , $\text{Var}(\hat{\beta}_1)$, and $\text{sd}(\hat{\beta}_1)$.

- (c) Based on the formula $\text{Var}(\hat{\beta}_1) = \sigma^2 / \text{SST}_x$, explain what happens to the precision of $\hat{\beta}_1$ when:

- (i) the error variance σ^2 increases;
- (ii) we add more observations that are spread out (not concentrated at $\bar{x}$);
- (iii) we add observations that are all equal to $\bar{x}$.

Block 3: From σ^2 to Standard Errors (10 min)

Estimating σ^2 and the t -Distribution

The formula $\text{Var}(\hat{\beta}_1) = \sigma^2 / \text{SST}_x$ involves $\sigma^2 = \text{Var}(u \mid \mathbf{x})$, which is **unknown**. We estimate it from the residuals:

$$\hat{\sigma}^2 = \frac{1}{n-2} \sum_{i=1}^n \hat{u}_i^2 = \frac{\text{SSR}}{n-2}$$

where $\hat{u}_i = y_i - \hat{y}_i$ are the OLS residuals and $\text{SSR} = \sum \hat{u}_i^2$ is the residual sum of squares. The divisor is $n - 2$ (not n) because we estimated two parameters ($\hat{\beta}_0$ and $\hat{\beta}_1$).

The **standard error** of $\hat{\beta}_1$ is the estimated standard deviation:

$$\text{SE}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\text{SST}_x}}$$

where $\hat{\sigma} = \sqrt{\hat{\sigma}^2}$ is the **residual standard error** reported in R.

Adding **SLR.6** (normality of errors), the t -statistic

$$t = \frac{\hat{\beta}_1 - \beta_1}{\text{SE}(\hat{\beta}_1)} \sim t_{n-2}$$

follows a t -distribution with $n - 2$ degrees of freedom. The t (not normal) arises because $\hat{\sigma}$ in the denominator is itself random, adding extra uncertainty.

The following R output will be used for Questions 2 and 3. An econometrician regressed y on x using 22 observations.

```
> model <- lm(y ~ x, data = mydata)
> summary(model)
...
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.2000     1.3000   4.000 0.000742 ***
x              2.4000     0.8000    A         B
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4 on 20 degrees of freedom

> confint(model)
              2.5 %    97.5 %
(Intercept)  2.488248 7.911752
x              C         D
```

Question 2 (Reading R output — variance and standard errors)

- (a) From the R output, identify $\hat{\sigma}$ (the residual standard error) and the number of observations n . Then compute $\hat{\sigma}^2$ and SST_x .

Hint: Use $SE(\hat{\beta}_1) = \hat{\sigma} / \sqrt{SST_x}$ and solve for SST_x .

- (b) Explain in one or two sentences why replacing σ with $\hat{\sigma}$ changes the distribution of the test statistic from standard normal to t .

- (c) Find A (the t -value) and B (the p -value) in the output. What null hypothesis does this t -statistic test?

Block 4: Hypothesis Testing (13 min)

Hypothesis Testing and Confidence Intervals

Two-sided test of $H_0: \beta_1 = \beta_{1,0}$ vs. $H_1: \beta_1 \neq \beta_{1,0}$:

- Compute $t = (\hat{\beta}_1 - \beta_{1,0})/\text{SE}(\hat{\beta}_1)$.
- Reject H_0 at significance level α if $|t| > t_{\alpha/2, n-2}$.
- **Equivalently:** reject H_0 if and only if $\beta_{1,0}$ falls **outside** the $(1 - \alpha)$ confidence interval.

One-sided test of $H_0: \beta_1 \leq \beta_{1,0}$ vs. $H_1: \beta_1 > \beta_{1,0}$:

- Same t -statistic.
- Reject H_0 at significance level α if $t > t_{\alpha, n-2}$.
- The critical value $t_{\alpha, n-2}$ is smaller than $t_{\alpha/2, n-2}$, so it is **easier to reject** in the direction of H_1 .

$(1 - \alpha)$ **Confidence interval:** $\hat{\beta}_1 \pm t_{\alpha/2, n-2} \cdot \text{SE}(\hat{\beta}_1)$.

Use the R output from Block 3. You may use the following critical values for the t_{20} distribution: $t_{0.025, 20} = 2.086$ and $t_{0.05, 20} = 1.725$.

Question 3 (Hypothesis testing and confidence intervals)

- (a) Using A and B from Question 2, test $H_0: \beta_1 = 0$ against $H_1: \beta_1 \neq 0$ at the 5% significance level. State your conclusion.

- (b) Find C and D (the 95% confidence interval for β_1). Show your work.

- (c) Using the confidence interval from (b), test $H_0: \beta_1 = 1$ against $H_1: \beta_1 \neq 1$ at the 5% significance level.

- (d) Now test $H_0: \beta_1 \leq 1$ against $H_1: \beta_1 > 1$ at the 5% significance level.

- (e) Parts (c) and (d) test closely related hypotheses about $\beta_1 = 1$, yet give opposite conclusions. Explain why.

Block 5: Type I/II Errors, Size, and Power (12 min)

Type I and Type II Errors

When we perform a hypothesis test, we make a decision (reject or fail to reject H_0) based on sample data. Since the data are random, we can make **mistakes**:

	H_0 is actually true	H_0 is actually false
Reject H_0	Type I Error	Correct (Power)
Fail to reject H_0	Correct	Type II Error

- **Type I Error (false positive):** Rejecting H_0 when H_0 is *true*. We conclude there is an effect, but in reality there is none.
- **Type II Error (false negative):** Failing to reject H_0 when H_0 is *false*. We miss a real effect.

Example — Courtroom trial: H_0 : defendant is innocent. Type I = convict an innocent person. Type II = acquit a guilty person.

Example — Drug trial: H_0 : drug has no effect. Type I = conclude the drug works when it doesn't. Type II = miss a drug that actually works.

Size = $P(\text{Type I Error}) = \alpha$. **Power** = $1 - P(\text{Type II Error})$.

Assume $\sigma_{\hat{\beta}}$ is **known**, so $Z = (\hat{\beta} - \beta_0)/\sigma_{\hat{\beta}} \sim N(0, 1)$ under H_0 .

Consider testing $H_0: \beta = 0$ vs. $H_1: \beta \neq 0$ at $\alpha = 0.05$, with $\sigma_{\hat{\beta}} = 2$.

Question 4 (Type I/II errors and power)

- (a) Define Type I and Type II error in the context of this test. Give a concrete interpretation if β measures the effect of a job training program on wages.

- (b) The test rejects H_0 when $|Z| > 1.96$. Show that the size of this test is exactly 0.05.

- (c) Compute the power of this test when the true value is $\beta^* = 6$.

Hint: Under $\beta^ = 6$, $Z = \hat{\beta}/2 \sim N(3, 1)$. Compute $P(|Z| > 1.96)$ using $W = Z - 3 \sim N(0, 1)$. You may use: $\Phi(1.04) = 0.851$.*

- (d) Compute the power when $\beta^* = 2$. Compare with part (c) and explain.

Hint: $\delta = 1$, so $Z \sim N(1, 1)$. You may use: $\Phi(0.96) = 0.831$.